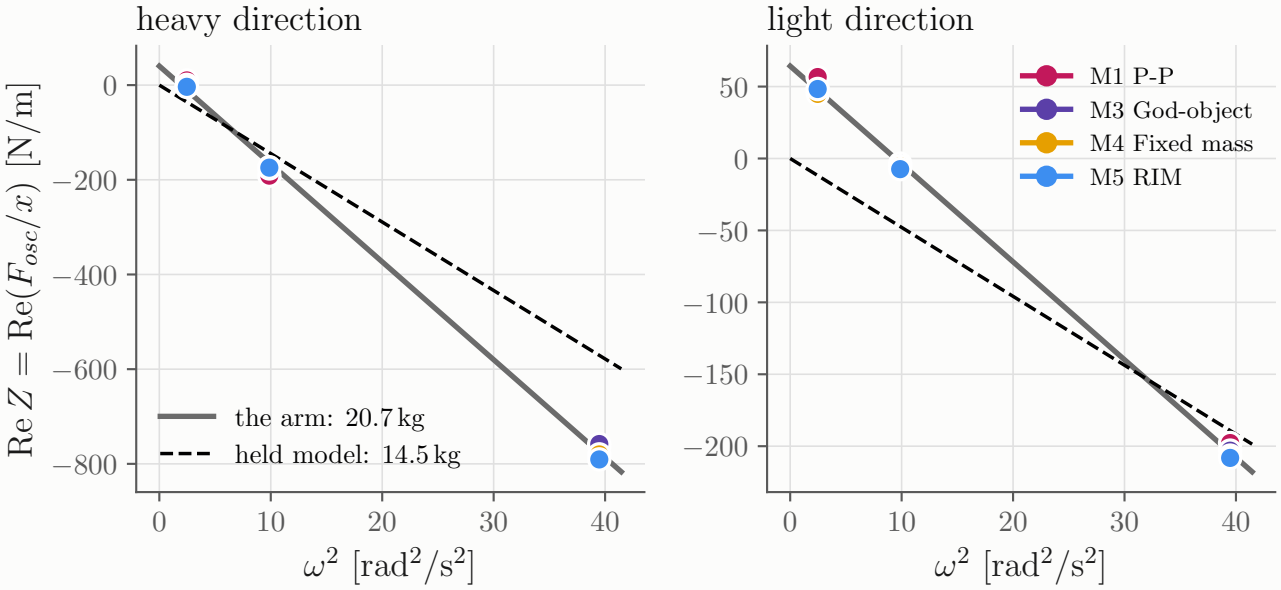

heavy direction
light direction
$\mathrm{Re}\, Z = \mathrm{Re}(F_{osc}/x)$ [N/m]
ω^2 [rad^2/s^2]
ω^2 [rad^2/s^2]
0
−200
−400
−600
−800
50
0
−50
−100
−150
−200
0
10
20
30
40
the arm: 20.7 kg
held model: 14.5 kg
M1 P-P
M3 God-object
M4 Fixed mass
M5 RIM